HW#3
REF: 3R's Lecture Slides and HP Instrument Lab Handout

1. List two examples of isotopic fission products that are present in the reactor core.

2. What effect on matter does ionizing radiation have that non-ionizing radiation does not? List two examples of each type of radiation (ionizing –vs- non-ionizing).

3. If you have a 1 Ci source of Cobalt-60, what is the exposure rate (R/hr) at 1 meter? Using the inverse square law ($DR_1 \cdot R_1^2 = DR_2 \cdot R_2^2$), what would the exposure rate be at 3 meters from the same source?

4. Radiation Units - Matching:

_____	Roentgen (R)	A.	Unit of Absorbed Dose (MeV/gram)
_____	RAD	B.	Activity in 1 gram of Radium = 3.7E10 DPS
_____	REM	C.	Unit of Exposure (coulombs/kg air)
_____	1 Becquerel (Bq)	D.	Unit of Biological Dose Equivalent (MeV /gram tissue)
_____	1 Curie (Ci)	E.	1 Decay Per Second (DPS)

5. Biological Dose – Matching:

_____	200 mREM/year	A.	LD 50/60
_____	100 mREM/year	B.	Average Annual Dose from Natural Background
_____	400 REM	C.	NRC Whole Body Annual Dose Limit for Radiation Workers
_____	360 mREM/year	D.	Average Annual Dose from Radon Exposure
_____	5 REM/year	E.	NRC Annual Dose Limit to Public from Licensed Activities

6. What type(s) of biological cells are at the greatest risk from radiation exposure?

7. List two <u>stochastic</u> and three <u>non-stochastic</u> biological radiation effects.

8. "ALARA" precepts dictate that, to keep doses <u>A</u>s <u>L</u>ow <u>A</u>s <u>R</u>easonably <u>A</u>chievable, you should do what four things (where possible)?

<u>HP Lab Questions:</u>

9. Fill in the answers for a Geiger-Muller detector –vs- an Ionization (proportional) Chamber:

	Geiger-Muller (GM)	**Ionization Chamber**
a) Operating Voltage Range		
b) Fill Gas(es)		
c) Radiation Types Detected		
d) Radiation Units used		

10. In a radioactive materials lab, what is the primary use of handheld GM detectors –vs- Ionization Chambers?

11. What type of neutron detector does our 'REM Ball' utilize? What is the function of the large polyethylene ball? What radiation units does the REM ball read out in?

12. What are two important pre-operational checks to verify before utilizing a hand-held radiation meter?